



CoBank Capstone Project

Building A Comprehensive Database For Credit Stress Analysis

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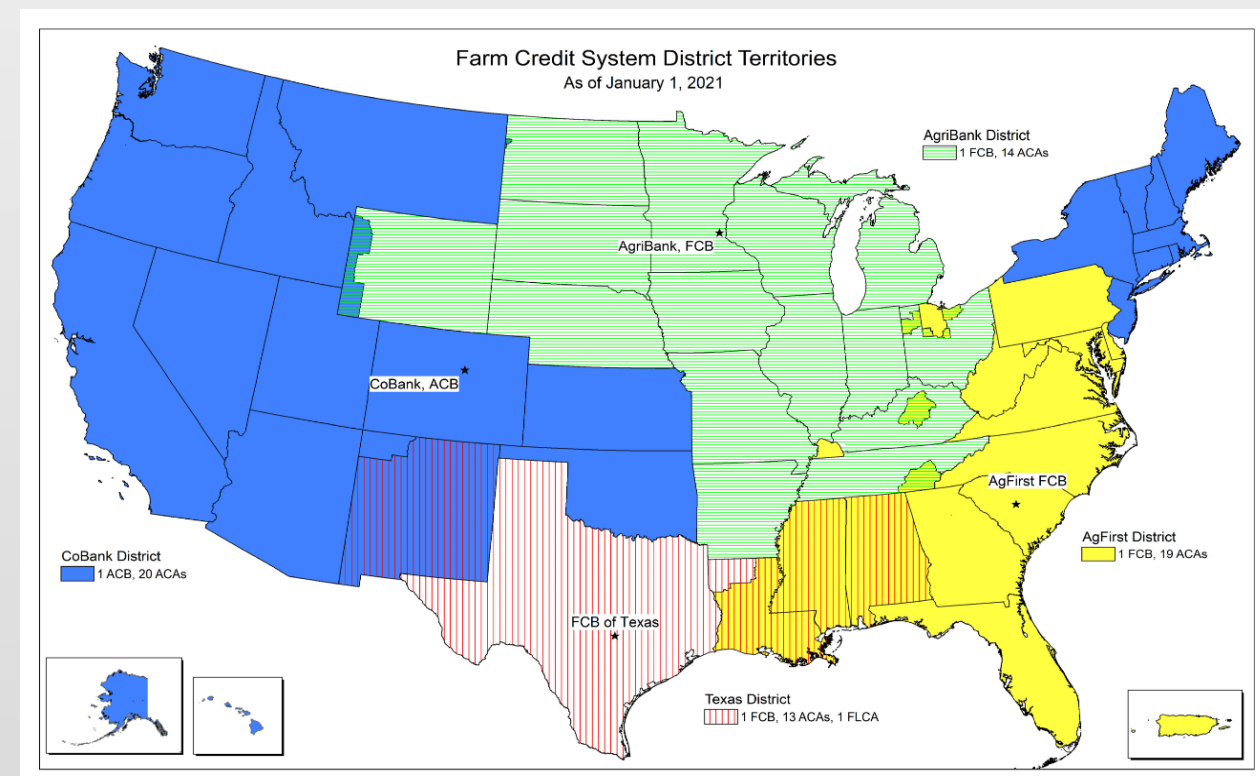
Company Overview

CoBank is the largest loan and financial service provider in the Farm Credit System, a national network of cooperatively owned financial institutions dedicated to supporting rural and agricultural communities throughout the United States.

CoBank's Enterprise Risk Management team wants our help to better understand the similarities and differences between farm creditors and commercial banks through data processing, visualizations, and econometric analysis to strengthen company risk management strategy.

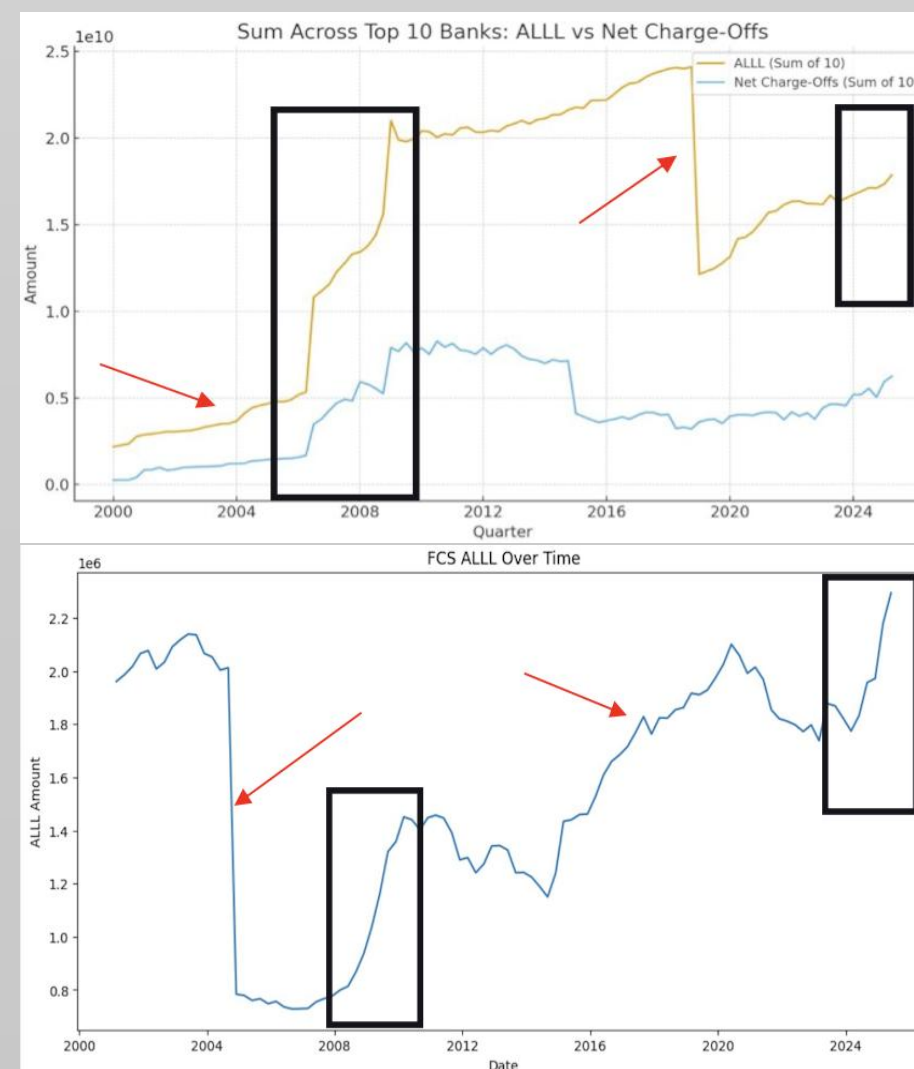
Project Advisors:

- Randy Medina (Vice President – Quantitative Risk Analytics)
- Eric Peter (Quantitative Risk Analyst)



Farm Credit System Database

Using publicly available FCS and commercial banking call reports, our team created a SQLite-based database containing balance sheet and credit stress information that allows data to be stored in one clean and query-able environment, saving CoBank an enormous amount of time and allowing for quick time series visualizations like the following trend visualizations of Allowance for Loan Losses and Net Charge-offs overtime:



Project Scope

Problem we address

Many stakeholders assume the Farm Credit System behaves like a typical commercial banking system, but CoBank needs evidence to confirm or challenge this view. At the same time, Farm Credit data is often fragmented and messy, making analysis slow and difficult without a unified, reusable database.

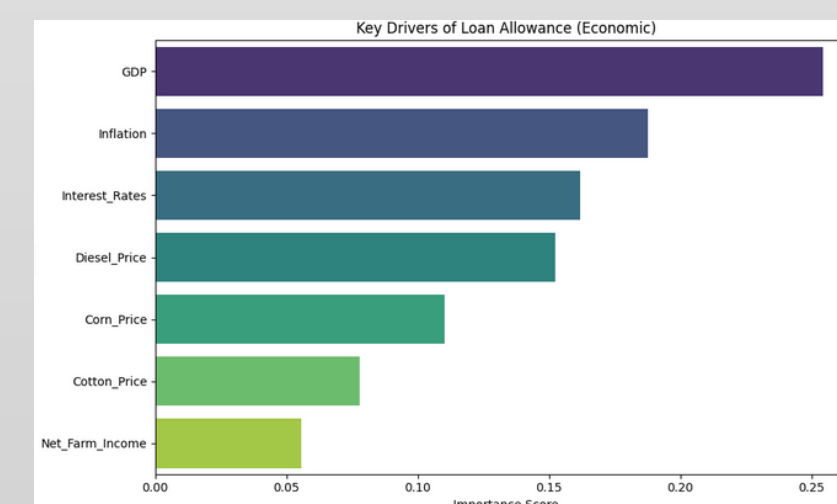
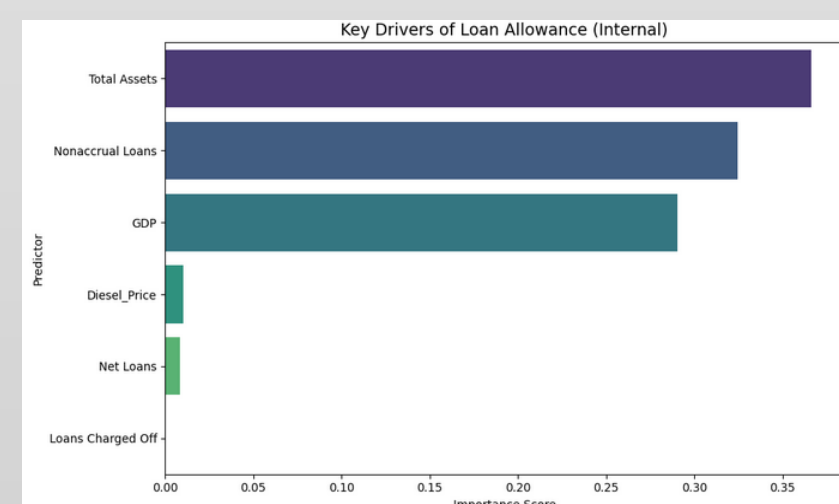
Objectives / Deliverables

- Build a reusable, well-documented data pipeline for Farm Credit System and commercial bank call-report data.
- Use this database to compare credit-stress behavior (allowances, net charge-offs, etc.) between the Farm Credit System and large commercial banks.
- Identify macroeconomic and agricultural indicators that drive credit stress in each system and prepare the foundation for forecasting and stress-testing tools to support CoBank's risk monitoring.

Total Loan Allowance Forecast

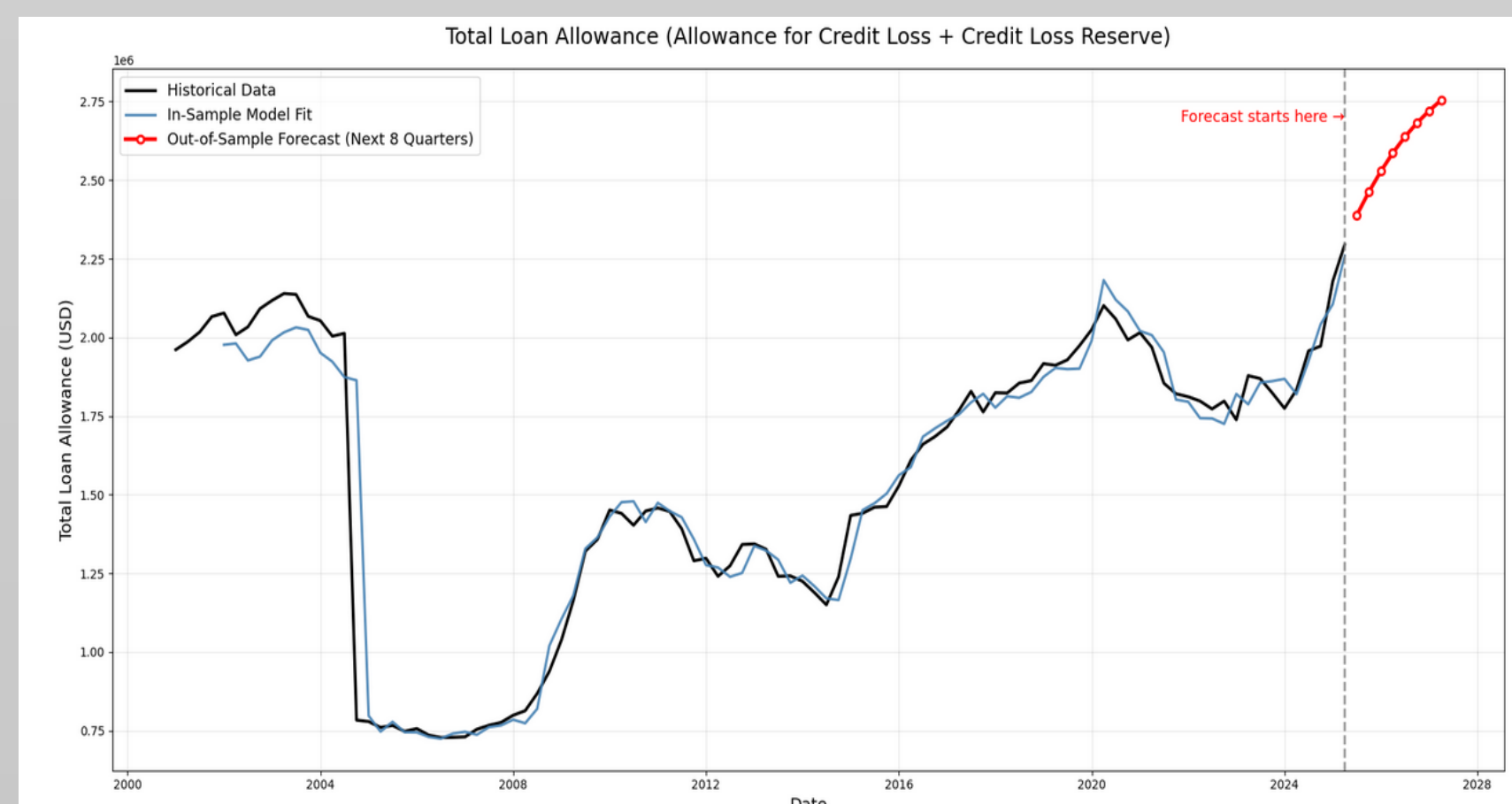
Random Forest Dimensionality Reduction

- Ranks variable importance by modeling hundreds of independent decision trees trained on random data subsets
- Ranked GDP, Nonaccrual Loans, Total Assets, and Diesel Prices as most important indicators of Total Loan Allowance



.Autoregressive Distributed Lag (ARDL) Regression Model

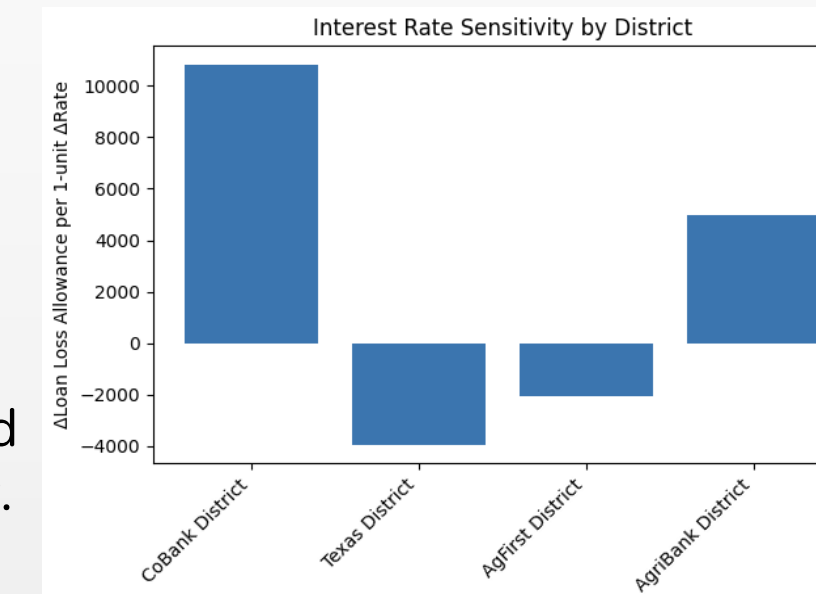
- Forecasts using past values of explanatory variables
- No significant multicollinearity detected by VIF test
- High Historical Accuracy (-130 BIC)



FCS Regional Credit Stress Behavior

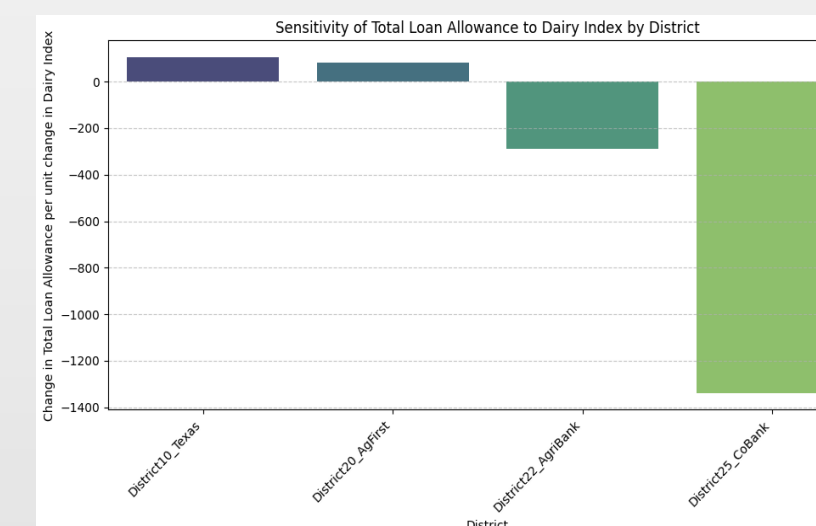
Macroeconomic Differences

- CoBank and AgriBank are highly sensitive to broad macroeconomic shocks, driving most systemwide movement..
- Texas and AgFirst display mild, contained sensitivities, with limited impact on overall system behavior.



Commodity Differences

- CoBank exhibits dominant commodity sensitivity across all sectors, far exceeding other districts.
- AgFirst and AgriBank show modest, mixed responses tied to their regional portfolio structures.
- Texas remains largely insensitive to commodity price shifts, reflecting low exposure in its lending mix.



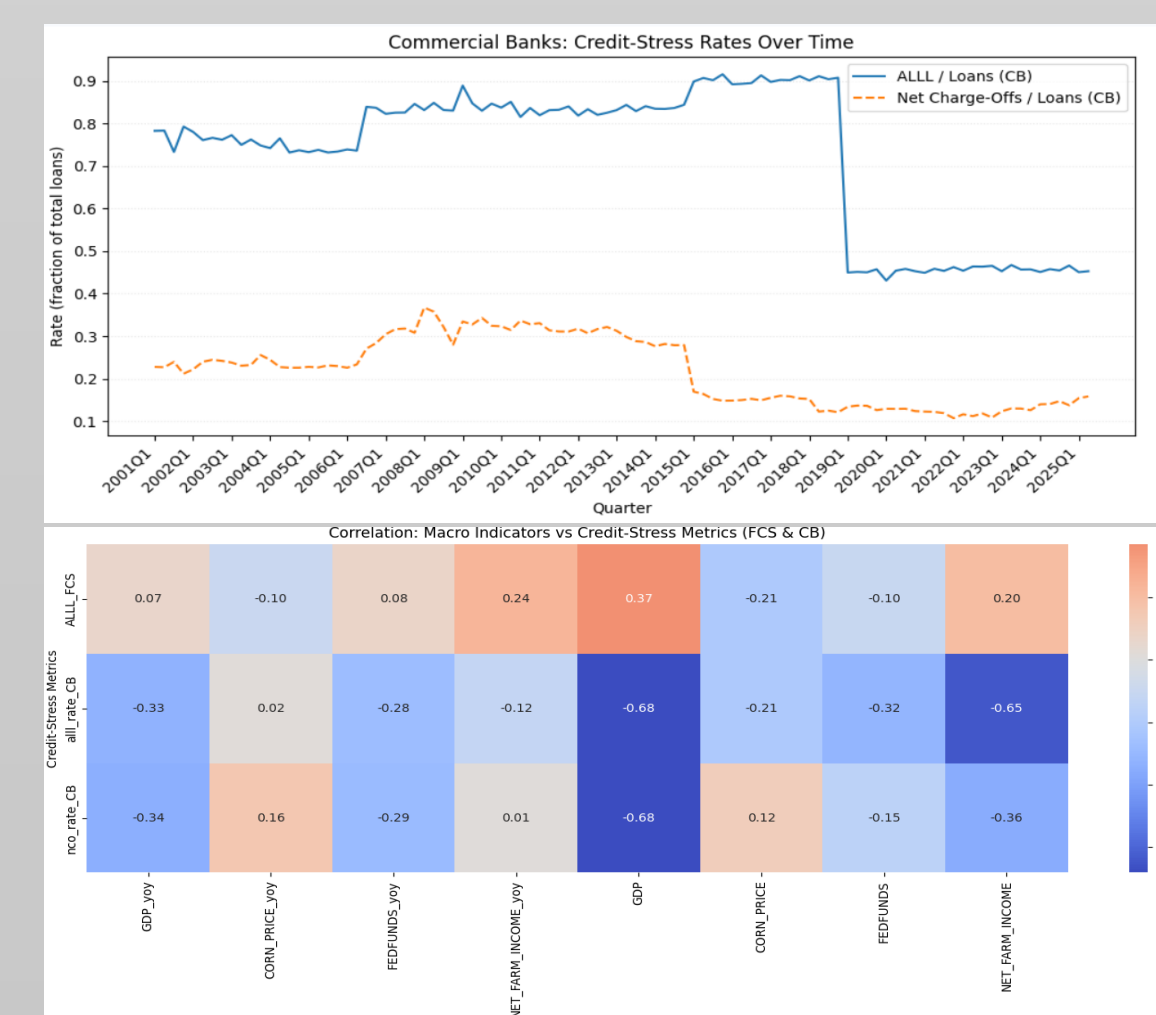
Unique and Shared Macro Drivers Between Systems

Unique Drivers

- Driven primarily by agricultural income cycles, not recessions.
- Strongly influenced by commodity price levels (diesel, corn, wheat)
- Credit conditions track Net Farm Income, a key profitability signal for borrowers.
- Impacted by weather-related yield shocks (droughts, floods)

Shared Macro Drivers

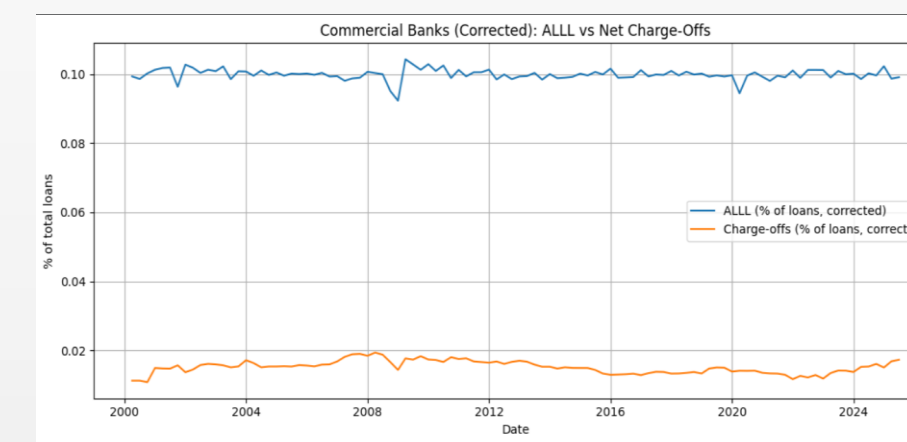
- Rising interest rates increases stress in both systems.
- Slowdowns in economic cycles affect credit quality across markets.
- Inflation & operating cost pressures influence borrower repayments
- National recessions: commercial banks react most intensely.



Charge-offs vs Non-accruals

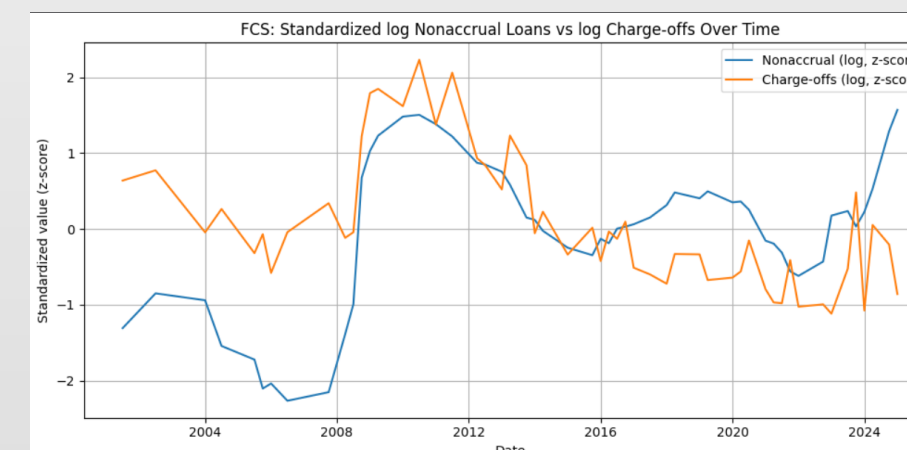
Commercial Banks (ALLL proxy)

- ALLL $\approx$ 0.10% of loans; charge-offs $\approx$ 0.02%.
- Move together over the cycle, but correlations are small.
- ALLL behaves like a slow-moving buffer, not a strong quarter-ahead predictor.



Farm Credit System (true nonaccruals)

- Linked nonaccrual loans with RIE2 charge-offs (cleaned and log-scaled).
- Positive relationship: higher nonaccruals $\leftrightarrow$ higher charge-offs.
- Nonaccruals spike slightly before charge-offs $\rightarrow$ useful early-warning signal.



Conclusion & Recommendations

Conclusions:

- The Farm Credit System and commercial banks respond to different economic forces, leading to distinct stress patterns
- FCS credit stress is more stable and agriculture-driven, with movements tied to Net Farm Income and commodity cycles.
- Commercial banks show higher volatility, reacting strongly to national recessions, GDP changes, and interest-rate shocks.
- FCS exhibits countercyclical characteristics relative to commercial banking, especially during major economic downturns.
- Non-accruals are a strong early-warning signal for commercial banks, while the relationship is weaker and delayed in FCS.

Recommendations:

- Develop early-warning dashboards tracking FCS and commercial bank divergence for ALLL, NCO rates, non-accruals.
- Use the unified database to support long-term forecasting, stress testing, and portfolio planning across both systems.

Acknowledgements

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